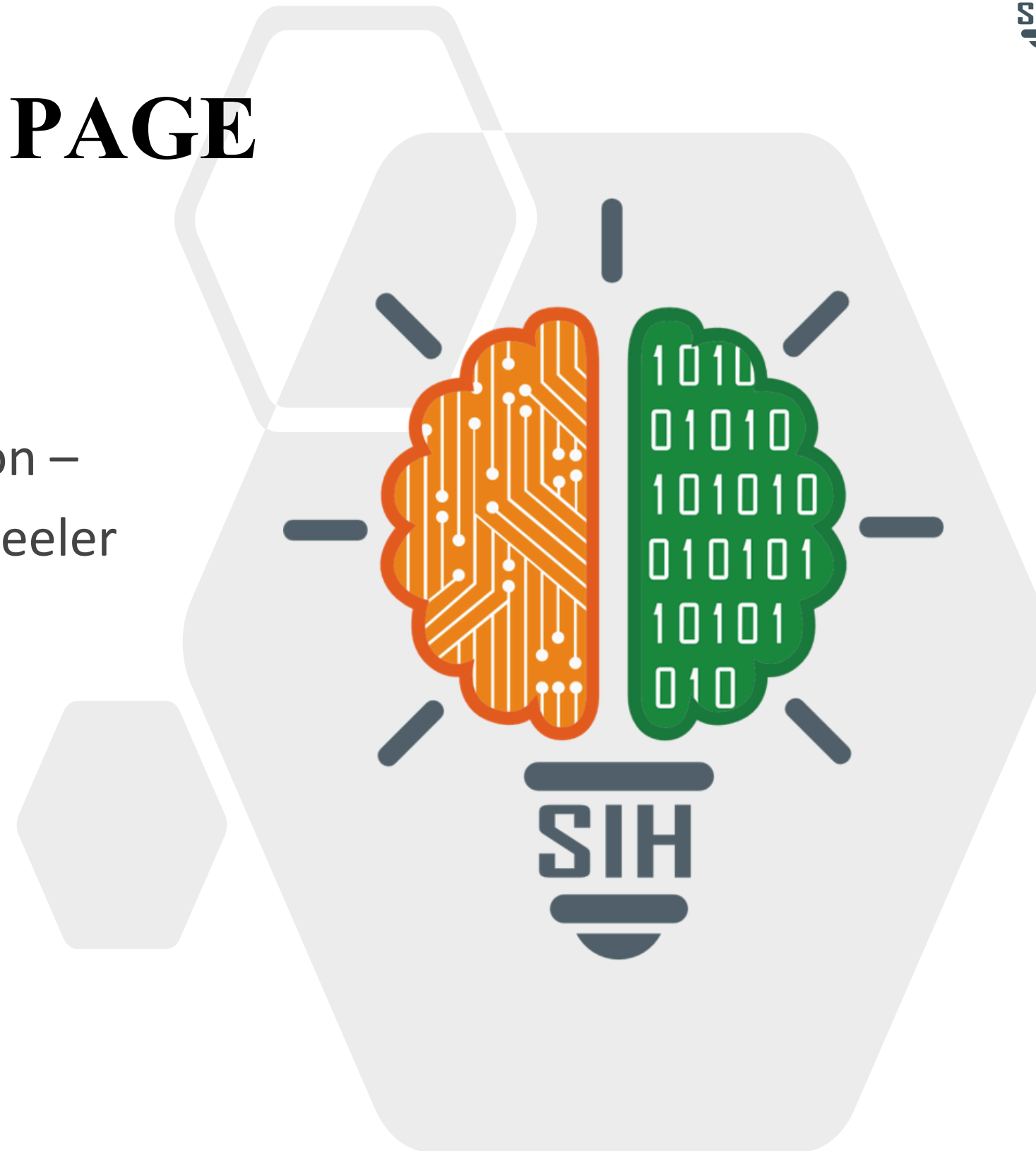


TITLE PAGE

- **PROBLEM STATEMENT ID** – SIH26203
- **PROBLEM STATEMENT TITLE-** Student Innovation –
Creating Intelligent Devices to Improve Two-Wheeler
Rider Safety and Emergency Response
- **THEME** - SMART VEHICLE
- **PS Category-** Software
- **Team ID-** SV-SW-07
- **Team Name** - Team MadCoders



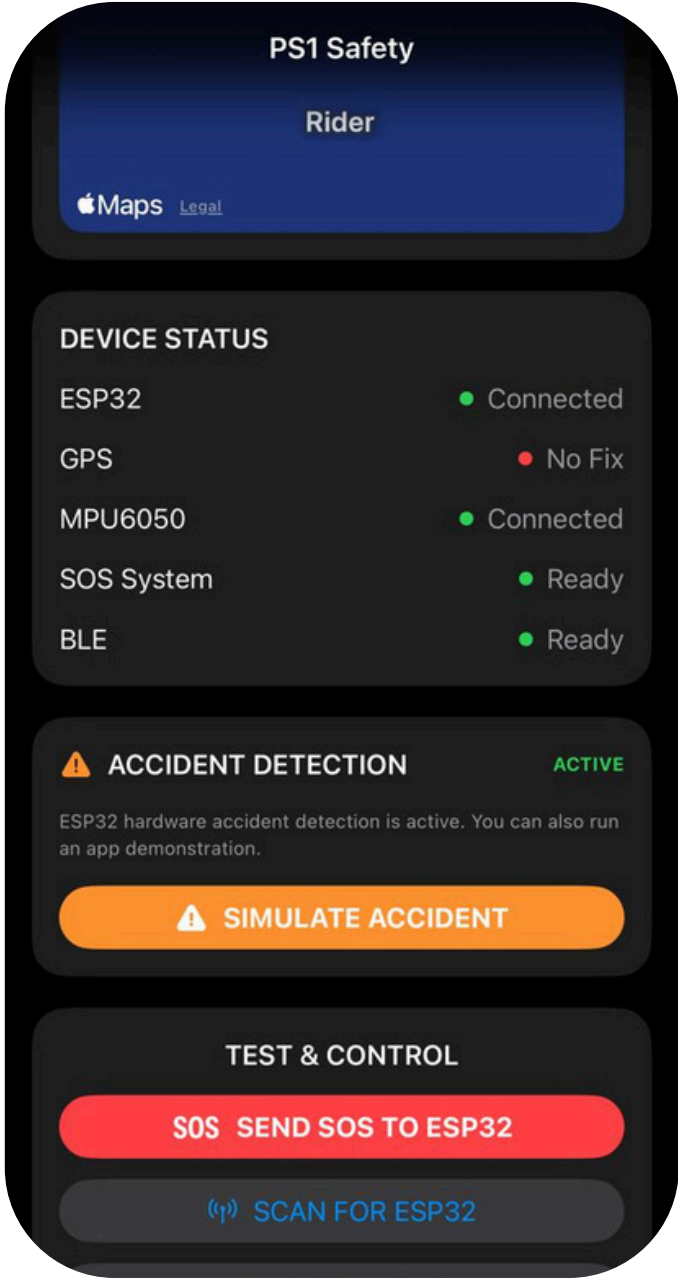
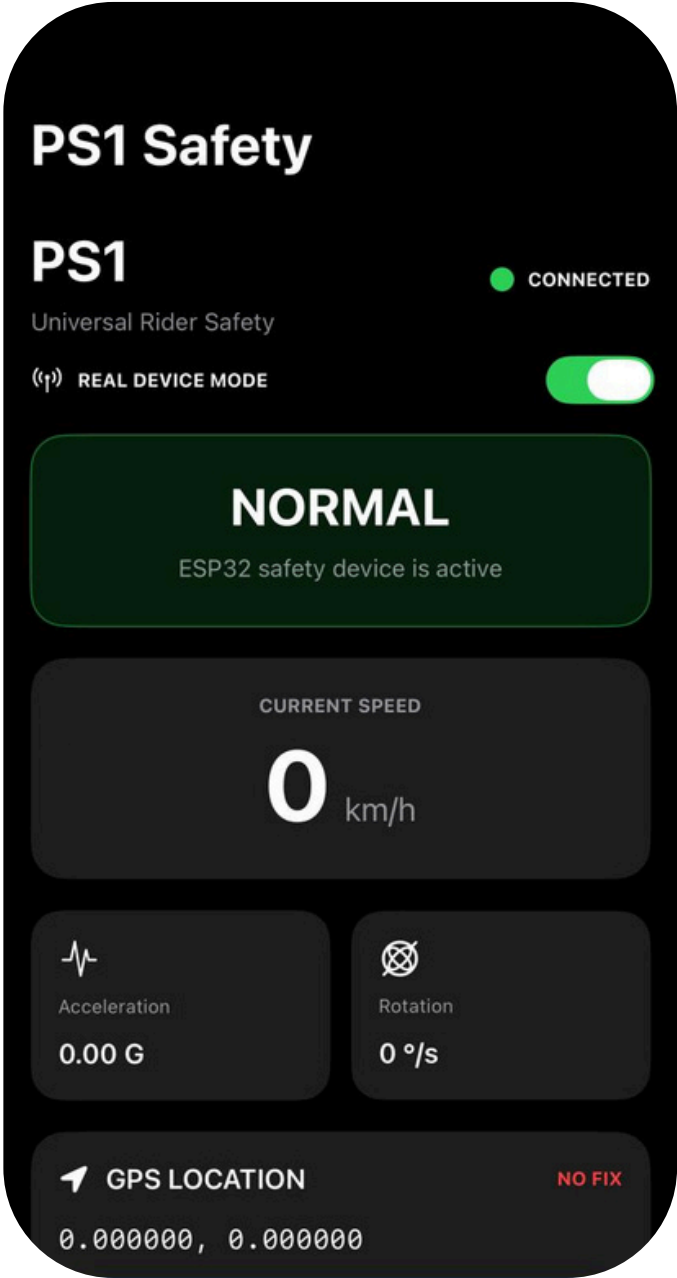
PROBLEM INTRODUCTION

Two-wheeler riders may face accidents, loss of control, personal threats, or impaired riding without timely help. Existing solutions usually address only one safety issue. A smart IoT system is needed to detect crashes, threats, and abnormal riding patterns, send SOS with live location to appropriate responders



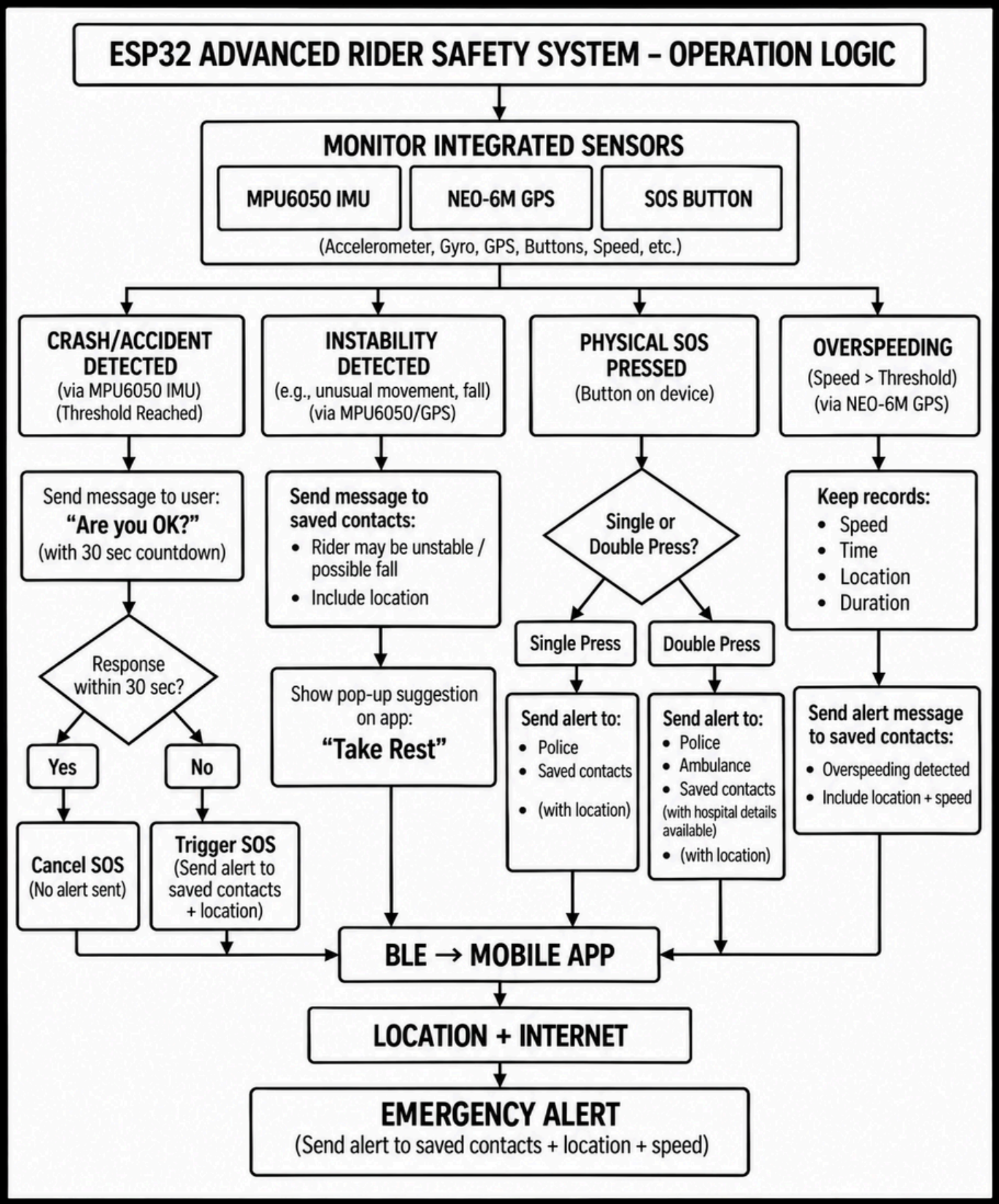
PROPOSED SOLUTION

- **Crash Detection**—Detects falls, impacts, and high vibrations.
- **Abnormal Riding Detection**—Identifies wobbling & unusual speed patterns.
- **Women’s SOS Button**—Instant emergency trigger on the handlebar.
- **Unified Safety System**—Combines automatic + manual emergency detection.
- **Smart SOS Alerts**—Sends emergency type + live GPS location to responders.
- **Mobile App**—Enables live tracking and quick false-alarm cancellation.



A RIDER MAY NEED EMERGENCY HELP BEFORE, DURING, OR WITHOUT A CONVENTIONAL CRASH.

TECHNICAL APPROACH



SENSING

- **MPU6050** - Motion / Impact
- **GPS** - Location / Speed
- **SOS Button** - Manual Trigger

PROCESSING

- ESP32
- Motion analysis
- Event classification
- False-alarm verification

COMMUNICATION

- BLE
- Android app
- GPS location
- Emergency contacts / response integration

D E T E C T → V E R I F Y → E S C A L A T E

REQUIRED SOFTWARE AND HARDWARE

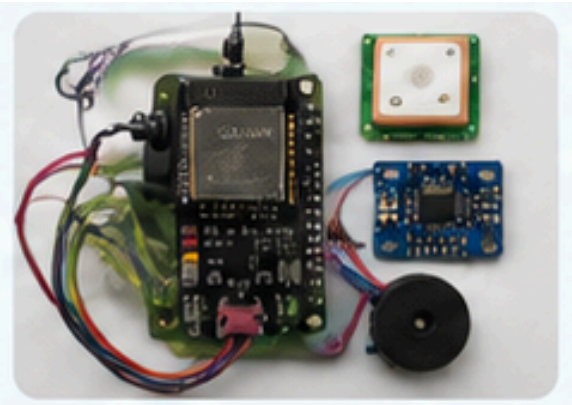
Software: Android App • BLE(bluetooth) • GPS/Maps • Accident Detection • Overspeed Detection • Emergency Alert System

Hardware: ESP32 DevKit • MPU6050 • NEO-6M GPS • Push Button • Buzzer • Breadboard • Jumper wires • LEDs/resistors • Power source/USB cable

FEASIBILITY AND VIABILITY

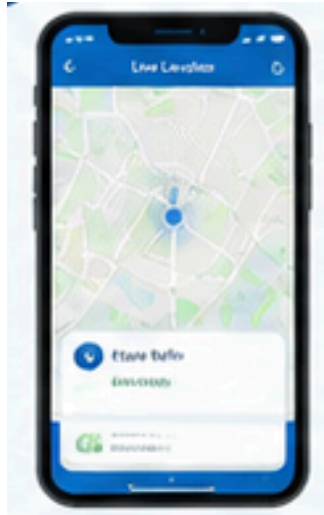
FEASIBILITY

01.HARDWARE



- ✓ ESP32 + IMU + GPS + BLE
- ✓ Compact retrofit architecture

02.SOFTWARE



- ✓ Android App
- ✓ BLE(bluetooth)
- ✓ GPS/Maps
- ✓ Tracking

03.COST

ESTIMATED PROTOTYPE COST

≤ ₹2,000

(Approximate)

*Based on listed components; excludes smartphone, enclosure and final installation

RISK AND MITIGATION

Detect Incident



Detect

System detects crash or abnormal event.

Verify

Rider verification (30 sec window).

Process & Analyse



Process

Sensor data is processed in real time.

Analyse

Check thresholds and validate event.

Send Alert



Send

Trigger emergency alert via mobile app.

Location

Attach real-time location and speed.

Notify Contacts



Notify

Send alert to saved contacts.

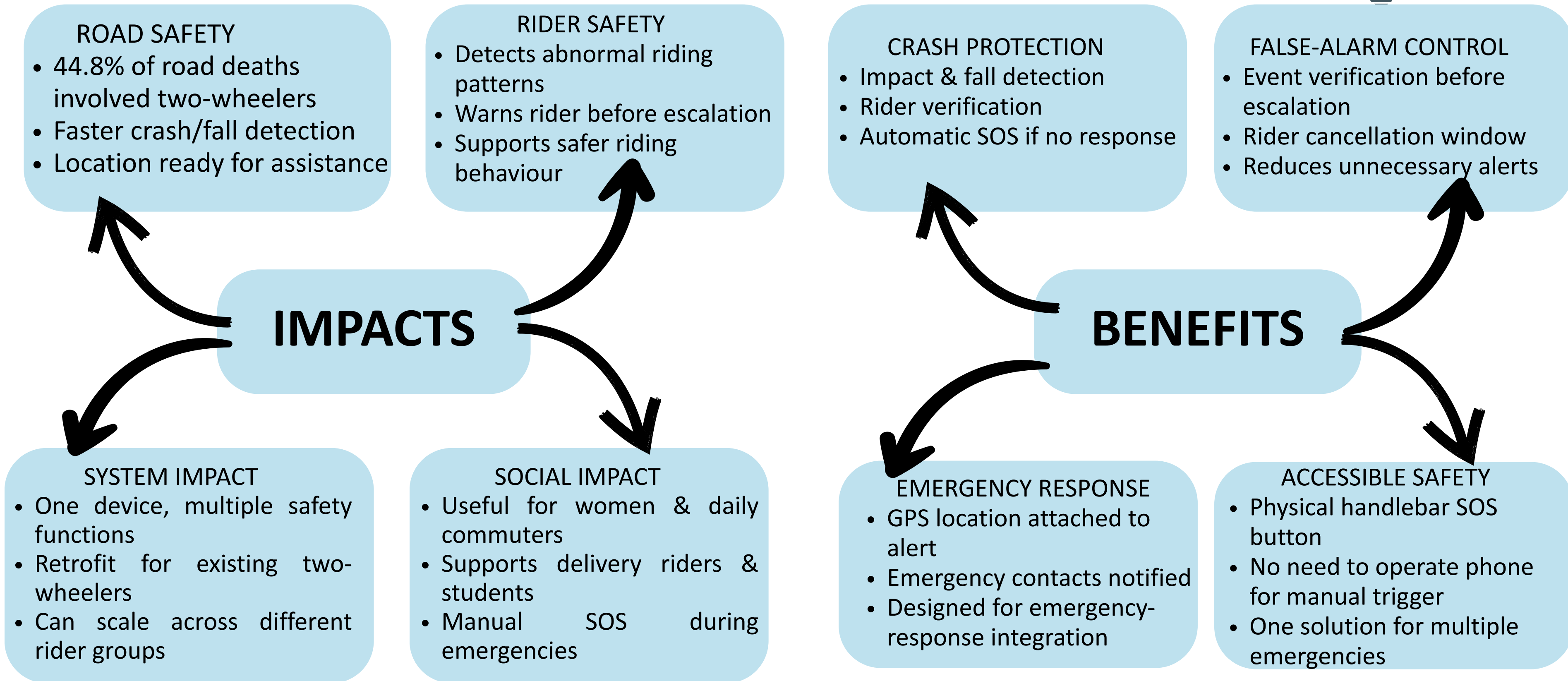
Contact

Also notify police/ambulance if required.

Component Price List

Item	Price (₹)
 Breadboard	₹50
 LEDs	₹6
 MPU6050	₹270
 GPS	₹320
 Male-to-Male Cables	₹15
 Female-to-Male Cables	₹15
 Female-to-Female Cables	₹15
 220Ω Resistors	₹10
 ESP32	₹380
 Pushbutton	₹3
 Buzzer	₹15

IMPACT AND BENEFITS



DETAILS / LINKS OF THE REFERENCE AND RESEARCH WORK

- Two-wheelers accounted for **44.8% of road-accident deaths** in India in **2023**.
- **54,568 people** were **killed** while not wearing helmets in 2023.
- WHO reports that quality helmets can reduce crash-death risk by more than 6x.
- Existing solutions often focus on one safety function such as GPS tracking or impact detection.
- **Future scope:** Extend the system for **women's safety**, including emergency alerts, location sharing and trusted-contact notification.

REFERENCE SOURCE:

- **Ministry of Road Transport & Highways:**
https://morth.gov.in/backend/documents/uploaded/1781177676_V1gUW8tJWT.pdf
- **World Health Organization:**<https://www.who.int/news-room/fact-sheets/detail/road-traffic-injuries>
- **HARDWARE PRODUCT LINKS-**<https://www.electropi.in/>

REFERENCE SOURCE:

- **DEMO Video of working prototype :-** https://drive.google.com/drive/folders/13Z8IZ_beVERgdy9IUmv5YKmLS7bMSPsX
- **Project Report of the Problem Statement :-** https://drive.google.com/drive/folders/1BNbUq5u_FJsoWqMExdggq6fmWhLAApO1